Fault-Tolerainzing at Constant Depth Overhead

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1 Local Decoders

Definition 1.1 (Locations)**.**

Definition 1.2 (DAG representation of quantum Gate)**.** Let U be a quantum gate, composed of local gates $U_1, \ldots, U_m$, and a set of locations V. We define the directed graph $G_U = (V, E)$ such that the vertices are associated with the locations, and for locations $x, y \in V$, there is an edge $(x, y) \in E$ if and only if one of the following holds:

1. x and y correspond to two consecutive time steps of the same qubit. Namely, y is the successor time location for x.
2. There is a local gate $U_i \in U$ that acts at locations x and y.

Definition 1.3 (Light Cone)**.** Let U be a quantum gate, and let q be a qubit on U output corresponds to the output location x_q. We say that a location x of U **affects** q if in G_U there is a directed path between x to q.

The **light cone** of q respecting to U is the union of all the locations of U that affect q. We denote it by $U_{[q]}$ When U is clear from the context, we will refer to the light cone respecting to it just as the **light cone of** q.

[COMMENT] Grammar

Definition 1.4 (Light Cones Intersections)**.** Let U be a quantum gate, and let q_1, q_2 be two qubits on its output. We define the **light cones intersections** of q_1 and q_2 to be all the locations which affect both q_1 and q_2. We denote it by $U_{[q_1,q_2]}$.

[COMMENT] Grammar

Definition 1.5 (U-configurations)**.** Let U be a quantum gate (not necessary unitary) , x_1 a location of U, and X_1 a single-qubit superoperator. Define $U[(x_1, X_1)]$ to be the quantum gate obtained from U by placing X_1 at location x_1 as follows:

1. If x_1 is a wait location, then we set X_1 to act at location x_1 instead.
2. If x_1 is a gate location, then we replace the gate with the gate which first applies X_1 and then applies the original gate.

For a set $X = \{(x_i, X_i)\}$ such that the x_i's are locations and the X_i's are single qubits superoperators. We denote by $U[X]$ the X-configuration of U defined to by placing each X_i at the x_i location. **[COMMENT]** Grammar

Definition 1.6 (Propgation)**.** Let U, X as in Definition 1.5, and let $Y = \{(y_i, Y_i)\}$ be a set of pairs composed by locations and superoperators. We say that $U[Y]$ is a propagation of $U[X]$ if $U[X] = U[Y]$.

Definition 1.7 (0/1-action of a Noise Channel)**.** A noise channel $\mathcal{N} : L(\mathcal{H}) \to L(\mathcal{H})$ is superoperator that maps quantum circuit U to a distribution over its possible configurations.

Fix an outcome of that distribution, namely a configuration $U[X]$, then:

1. We define the 0-action of $\mathcal{N}$ to be all locations in $U[X]$ that deviate from the locations on U, which are all the locations in X.

2. On the other hand, we define the 1-action of $\mathcal{N}$ to be a set of locations, such for each location there exists at least one configuration $Y = \{(y_i, Y_i)\}$ such that Y contains it and $U[Y]$ is a propagation of $U[X]$.

   [COMMENT] Grammar

Definition 1.8 (0-Clean/Dirty Locations)**.** Let S be a subset of locations, and let $\mathcal{N}$ a noise channel acting on the quantum circuit. We define the event S **is** 0**-clean** to refer to all configurations in which no location in S belongs to the 0-action of $\mathcal{N}$.

Similarly, We define the event S **is** 0**-dirty** to refer to all configurations for which at least one qubit in S belongs to the 0-action of $\mathcal{N}$.

When, S is single a location associated with a qubit or a classical bit q, and the location is clear form the context, we will abuse the notation and use q **is** 0**-clean/dirty** to refer the event describing the whether or not the location is in the 0-action.

In addition, we will use the notations to refer events describing light cones, for example, for a quantum gate U and two qubits q_1 and q_2 the event **Their light cones intersections is** 0**-clean** refer to all configurations in which the subset of location in their intersection is 0-clean.
[COMMENT] Grammar

Remark 1.1. We add the digit 0 at the beginning of the status to emphasis that the notation describes the influence of the channel independent to the computation defines by U. In particular, if location y is in the light cone of x, and y is 0-dirty, then it doesn't implies that x is also 0-dirty. **[COMMENT]** Grammar

Definition 1.9 (1-Clean/Dirty Location)**.** Using the same notations of Definition 1.8. We say that a location x is 1-dirty if it belongs to the 1-action of $\mathcal{N}$, and 1-clean otherwise. **[COMMENT]** Grammar

Definition 1.10 (p-Local Stochastic Noise)**.** Let U be a quantum gate with a set of locations $\mathcal{L}$, we say that a channel $\mathcal{N}$ is a **local stochastic noise** channel if there exist a constant $p \in (0,1)$ such that $\mathcal{N}(U)$ defines a probability space in which for any $S \subset \mathcal{L}$, the probability that all the qubits in S are 0-dirty is upper bounded by $p^{|S|}$. **[COMMENT]** Grammar

Definition 1.11 (p-Depolarized Noise)**.** Let U be a quantum gate with a set of locations $\mathcal{L}$, for a constant $p \in (0,1)$ we say that a channel $\mathcal{N}$ is a p**-depolarized noise** channel, if the event that a location x is 0-dirty is distributed by $\mathbf{Ber}(p)$. In particular, the location are distributed independently. **[COMMENT]** Grammar

Definition 1.12. Let C be a quantum error correction code, let D be a quantum gate, and let $\mathcal{N}$ be a $2p(1-p)$-depolarizing channel (local stochastic). Consider the following setting: D is fed by a quantum state encoded in C through input locations. $\mathcal{N}$ is applied only on the input locations and on the inter locations of D; in particular, the output locations are not affected by $\mathcal{N}$. Then C is a local decoder for C if, under that setting, the following holds:

1. **Local improvement.** There exists function $M : (0,1) \to (0,1)$ such that the probability of any output location q being 1-dirty is upper bounded by

$$\mathbf{Pr}\left[q_1 \text{ is 1-dirty}\right] = M(2p(1-p)) < p^{\frac{100}{99}}$$

2. **Locality computation.** There is a constant l such that for any qubit q, there are at most l location in its light cone $|D_{[q]}| \leq l$, and in addition, any location is contained in the light cone of at most l qubits.

3. 0**-clean intersection implies 'independence'.** For any subsets of qubits $q_1, q_2, .., q_k$ on D output, if the light cones intersection of any two of them $D_{[q_1,q_2]}$ is 0-clean, Then the probability that they all belong to 1-action, i.e not corrected or being faulty, is bounded as follow:

$$\begin{aligned}&\mathbf{Pr}\left[q_1, q_2, .., q_k \text{ are 1-dirty} | \bigcap_{q_1' \neq q_2' \in q_1,..,q_k} D_{[q_1',q_2']} \text{ is 0-clean}\right] \\ &\leq \prod_{q_i \in q_1,..,q_k} \mathbf{Pr}\left[q_1 \text{ is 1-dirty}\right]\end{aligned}$$

We require one more additional requirement - **Logical computation.** If we denote by σ and ρ the input and the output states, and σ is a codeword of C, then an ideal running of D over σ results in an encoded codeword, namely ρ is also a codeword.

For example, if we take k large enough and consider the (n,k)-Toric, then both the swift-rule and the flip upon majority are local decoders.

[COMMENT] Grammar

Lemma 1.1. Consider a code C with a local decoder D, and let σ and ρ be n-qubit quantum states such that under an ideal running $\rho = D(\sigma)$. If σ is a codeword of C, then there exists a constant $c > 0$, such that for any bounded norm Hermitian operator X, it holds that:

$$\mathbf{Tr} X \left(\mathcal{N}_p \circ D\right)\left(\mathcal{N}_p \sigma\right) \leq \mathbf{Tr} X \mathcal{N}_p\left(\rho\right) + 2||X||e^{-\Theta(\sqrt{n})}$$

Claim 1.1. Consider the running of $\left(\mathcal{N}_p \circ D\right)\left(\mathcal{N}_p \sigma\right)$, using the notations in Lemma 1.1. Let S be a subset of qubits chosen uniformly random [1]. And let J be a subset of S such that for any $q \in J$ there is another qubit $q' \neq q$ in J, such $D_{[q,q']}$ is 0-dirty. Then:

$$\mathbf{Pr}\left[|J| \geq \frac{1}{100}|S|\right] \leq 2e^{-c\sqrt{n}}$$

Proof. Let S be an arbitrary subsets of qubits. For each qubit q in S define the indicator X_q that gets 1 if there is $q' \neq q$ in S such $D_{[q,q']}$ is 0-dirty. So $J = \sum_{q \in S} X_q$ and in expectation we have:

$$\begin{aligned}&\mathbf{E}_{\sim \mathcal{N}_p}\left[|J|\right] = \sum_{q \in S} \mathbf{E}_{\sim \mathcal{N}_p}\left[X_q\right] \leq \sum_{q \in S}\left(|D_{[q]}| \times \overbrace{2}^{\text{input locations}}\right) p \\ &\leq 2pl|S|\end{aligned}$$

Now we are about to use the MacDiarmid's inequality, stated as follow:

Theorem 1.1 (MacDiarmid's inequality)**.** For independent random variables $X_1, X_2, \ldots, X_n$ taking values in a set A, and for any function $f : A^n \to \mathbb{R}$ satisfying the bounded differences condition:

$$|f(x) - f(x')| \leq c_i$$

whenever the vectors x and x' differ only in the i-th coordinate, the inequality is given by:

$$\begin{aligned}\mathbf{Pr}\left[f(X_1, X_2, \ldots, X_n) - \mathbf{E}\left[f(X_1, X_2, \ldots, X_n)\right] \geq t\right] \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right) \\ \mathbf{Pr}\left[f(X_1, X_2, \ldots, X_n) - \mathbf{E}\left[f(X_1, X_2, \ldots, X_n)\right] \leq -t\right] \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right)\end{aligned}$$

for all $t > 0$.

[1] each qubit is taken with probability $\frac{1}{2}$

[COMMENT] Right only in the independent case. For a fixed S, we have that $|J|$ is a function of the 0-action in the running of $(\mathcal{N}_p \circ D)(\mathcal{N}_p\sigma)$. Furthermore, N_p is a depolarized channel. We have that 0-dirty locations are distributed independently. Lastly, since D is a local decoder Definition 1.12, each location is contained in the light cone of at most l qubits. We have that adding or removing a single location from a given assignment of 0-action can differ $|J|$ by at most l.

Therefore, the probability that $|J|$ is $|S|^{\frac{3}{4}}$ away from the expected is bounded by:

$$\mathbf{Pr}_{\sim\mathcal{N}_p}\left[|J| - \mathbf{E}\left[|J|\right] \geq |S|^{\frac{3}{4}}\right] \leq e^{-2\frac{|S|^{\frac{3}{2}}}{\Sigma_{\text{locations}} l^2}} \leq e^{-2\frac{|S|^{\frac{3}{2}}}{n(l+1)l^2}}$$

In particular, for any $\alpha \in (0,1)$ we have that for any $|S| > \alpha n$:

$$\mathbf{Pr}_{\sim\mathcal{N}_p}\left[|J| - \mathbf{E}\left[|J|\right] \geq |S|^{\frac{3}{4}}\right] \leq e^{-\frac{2\alpha^{\frac{2}{3}}}{(l+1)l^2}\sqrt{n}}$$

Thus for small enough $p \in (0,1)$, with probability greater than $1 - 2e^{-\frac{2\alpha^{\frac{2}{3}}}{(l+1)l^2}\sqrt{n}}$, the number of dependent bits is less than:

$$\left(2p(1-p) + \frac{1-(1-4p(1-p))^{\Delta_2}}{2}\right)\Delta^l|S| + |S|^{\frac{3}{4}} \to \frac{|S|}{100}$$

□

Claim 1.2. Consider the notations in Claim 1.1, denote by K the output locations in the 1-action, namely the qubits which at the end of the running deviate from the result of the ideal running. Then:

$$\mathbf{Pr}_{\mathcal{N}_p}\left[|K| \geq \frac{1}{2}p^l n\right] \geq 1 - e^{-\frac{2\left(\frac{1}{2}p^l\right)^2 n}{(l+1)l^2}}$$

Proof. For any qubit, denote by X_q the indicator that indicates that the output location associated with q is 1-dirty. The probability that $X_q = 1$ is greater than the probability that all of the locations in $D_{[q]}$ belong to the 0-action [2]. Thus:

$$\begin{aligned}\mathbf{E}_{\sim\mathcal{N}_p}\left[|K|\right] &= \sum_{q\in K}\mathbf{E}_{\sim\mathcal{N}_p}\left[X_q\right] \geq \sum_{q\in K} p^{\left(|D_{[q]}|\times 2\right)} \\ &\geq np^{2l}\end{aligned}$$

Using again MacDiarmid's inequality we have that:

$$\mathbf{Pr}_{\sim\mathcal{N}_p}\left[|K| \leq \frac{1}{2}np^{2l}\right] \leq \mathbf{Pr}_{\sim\mathcal{N}_p}\left[|K| - \mathbf{E}\left[|K|\right] \leq -\frac{1}{2}np^{2l}\right] \leq e^{-\frac{2\left(\frac{1}{2}p^l\right)^2 n}{(l+1)l^2}}$$

□

[2] If the code is not trivial

Corollary 1.1. By Claim 1.2 and Claim 1.1, we get that

$$\sum_{S\subset[n]} \mathbf{Pr}_{\sim\mathcal{N}_p}\left[S=K\right] \leq \sum_{\substack{S\subset[n]\\ |S|\geq\frac{1}{2}p^l n}} \mathbf{Pr}_{\sim\mathcal{N}_p}\left[S=K\right] + e^{-\frac{2\left(\frac{1}{2}p^l\right)^2 n}{(l+1)l^2}}$$

$$\leq \sum_{\substack{S\subset[n]\\ |S|\geq\frac{1}{2}p^l n}} \mathbf{Pr}_{\sim\mathcal{N}_p}\left[S=K \middle| |J|-\mathbf{E}\left[|J|\right]\leq |S|^{\frac{3}{4}}\right] + e^{-\frac{2\left(\frac{1}{2}p^l\right)^2 n}{(l+1)l^2}} + e^{-\frac{2\left(\frac{1}{2}p^l\right)^{\frac{2}{3}}}{(l+1)l^2}\sqrt{n}}$$

$$\leq \sum_{\substack{S\subset[n]\\ |S|\geq\frac{1}{2}p^l n}} \mathbf{Pr}_{\sim\mathcal{N}_p}\left[S=K \middle| |J|-\mathbf{E}\left[|J|\right]\leq |S|^{\frac{3}{4}}\right] + 2e^{-\frac{2\left(\frac{1}{2}p^l\right)^{\frac{2}{3}}}{(l+1)l^2}\sqrt{n}}$$

In particular, where $|S|\geq\frac{1}{2}p^l n$, we have that for small enough p and large n:

$$\mathbf{E}\left[|J|\right]+|S|^{\frac{3}{4}} \leq 2pl|S|+|S|^{\frac{3}{4}} \leq 2pl|S|+n^{\frac{3}{4}} \leq 2pl|S|+\frac{1}{400}p^l n \leq \frac{1}{100}|S|$$

Hence we get that:

$$\sum_{S\subset[n]} \mathbf{Pr}_{\sim\mathcal{N}_p}\left[S=K\right] \leq \sum_{\substack{S\subset[n]\\ |S|\geq\frac{1}{2}p^l n}} \mathbf{Pr}_{\sim\mathcal{N}_p}\left[S=K \middle| |J| \leq \frac{1}{100}|S|\right] + 2e^{-\frac{2\left(\frac{1}{2}p^l\right)^{\frac{2}{3}}}{(l+1)l^2}\sqrt{n}}$$

Claim 1.3. For a state ρ subjected to a q-local stochastic noise, we have that the probability to that the portion of the faulty bits is greater than $\alpha\in(0,1)$ decay exponentially in n.

Proof. First use the Markov inequality to bound the space that $\geq n\alpha$ so we get $\geq \frac{1}{2}\cdot\frac{1}{\alpha}\cdot 2^n$, and then use the local stochastic property:

$$\leq C\frac{\frac{1}{2}}{\alpha}\cdot 2^n\cdot q^{\alpha n} = \frac{\alpha}{2}C\left(2q^{\alpha}\right)^n$$

For big enough α we see that the probability decay exponentially. (An also a way to prove that $q^q\geq\frac{1}{2}$).

□

Fix a subset of qubits $S\subseteq n$. Denote by J the qubits in S such that they share a 0-dirty light cone intersection with another qubit in S. Let's denote by $Y\subseteq D_{[S]}$ the subset of locations in the light cone of S which are 0-dirty. Observe that $|J|\leq|Y|\cdot l$, because any faulty location is in at most l light cones of output locations. Thus:

$$\mathbf{Pr}_{\sim\mathcal{N}_p}\left[|J|\leq\frac{|S|}{100}\right] = 1-\mathbf{Pr}_{\sim\mathcal{N}_p}\left[|J|>\frac{1}{100}|S|\cdot l\right]$$
$$\geq 1-\mathbf{Pr}_{\sim\mathcal{N}_p}\left[|Y|\cdot l>\frac{1}{100}|S|\right]$$
$$\geq 1-\mathbf{Pr}_{\sim\mathcal{N}_p}\left[|Y|>\frac{1}{100l^2}|D_{[S]}|\right]$$
$$\geq 1-\frac{1}{100l^2}\left(2p^{\frac{1}{100l^2}}\right)^{|D_{[S]}|}$$
$$\geq 1-\frac{1}{100l^2}\left(2p^{\frac{1}{100l^2}}\right)^{|S|l}$$

Lemma Proof:

Proof. For a subset of qubits $S \subset [n]$, let ρ^S be the density matrix obtained from $\rho = (\mathcal{N}_p \circ D)(\mathcal{N}_p \sigma)$ conditioned on the event that the **[COMMENT]** Should define it. output 1-action lies on S. Thus, ρ can be written as:

$$\rho = \sum_{S \subset [n]} \rho^S \mathbf{Pr}_{\sim \mathcal{N}_p}[S = K]$$

Therefore:

$$\begin{aligned}
&\mathbf{Tr} X (\mathcal{N}_p \circ D)(\mathcal{N}_p \sigma) = \sum_{S \subset [n]} \mathbf{Tr}\left(X \rho^S\right) \mathbf{Pr}_{\sim \mathcal{N}_p}[S = K] \\
&\leq \sum_{\substack{S \subset [n] \\ |S| \geq \frac{1}{2} p^l n}} \mathbf{Tr}\left(X \rho^S\right) \mathbf{Pr}_{\sim \mathcal{N}_p}\left[S = K \Big| |J| \leq \frac{1}{100}|S|\right] \\
&\leq \sum_{\substack{S \subset [n] \\ |S| \geq \frac{1}{2} p^l n}} \mathbf{Tr}\left(X \rho^S\right) M(2p(1-p))^{\frac{99}{100}|S|} + 2||X||e^{-\frac{2\left(\frac{1}{2}p^l\right)^{\frac{2}{3}}}{(l+1)l^2}\sqrt{n}} \\
&\leq \sum_{\substack{S \subset [n] \\ |S| \geq \frac{1}{2} p^l n}} \mathbf{Tr}\left(X \rho^S\right) \left(p^{\frac{100}{99}}\right)^{\frac{99}{100}|S|} + 2||X||e^{-\frac{2\left(\frac{1}{2}p^l\right)^{\frac{2}{3}}}{(l+1)l^2}\sqrt{n}} \\
&\leq \sum_{\substack{S \subset [n] \\ |S| \geq \frac{1}{2} p^l n}} \mathbf{Tr}\left(X \rho^S\right) p^{|S|} + 2||X||e^{-\frac{2\left(\frac{1}{2}p^l\right)^{\frac{2}{3}}}{(l+1)l^2}\sqrt{n}} \\
&\leq \mathbf{Tr} X (\mathcal{N}_p \circ D) + 2||X||e^{-\frac{2\left(\frac{1}{2}p^l\right)^{\frac{2}{3}}}{(l+1)l^2}\sqrt{n}}
\end{aligned}$$

□

Definition 1.13. Consider the moment after the application of the depolarizing channel, and right before the correction by the decoder.

For a subset of qubits S, we say that a qubit $x \in S$ is depended if there exists $y \in S$ such:

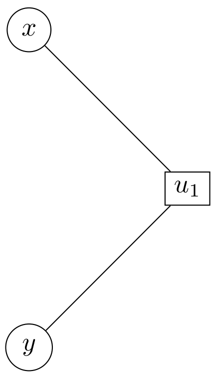
1. Either x and y share a check that is not satisfied. (Type I)
2. Or, x and y have checks which share a bit that was flipped by the application of the channel. (Type II)

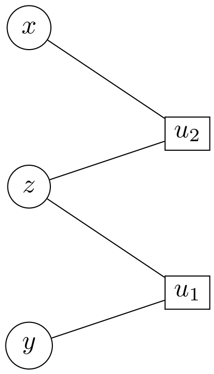
Dependence Type I

$$\begin{aligned}
&\mathbf{Pr}[(x, y) \in e | u_1 \text{ is satisfied}] \\
&= \mathbf{Pr}[x \in e | u_1 \text{ is satisfied}] \mathbf{Pr}[y \in e | u_1 \text{ is satisfied}] \\
&\leq \mathbf{Pr}[x \in e] \mathbf{Pr}[y \in e]
\end{aligned}$$

Dependence Type II

$$\begin{aligned}
&\mathbf{Pr}[(x, y) \in e | z \text{ is not flipped}] \\
&= \mathbf{Pr}[x \in e | z \text{ is not flipped}] \mathbf{Pr}[y \in e | z \text{ is not flipped}] \\
&\leq \mathbf{Pr}[x \in e] \mathbf{Pr}[y \in e]
\end{aligned}$$





2 Memory

Definition 2.1. Let $\rho^{(0)}$ be a mixed state over codewords. Define:

$$\rho^{(0)} = \rho$$
$$\rho^{(t)} = D \circ \mathcal{N}_p \rho^{(t-1)}$$

Theorem 2.1. There is $p_{\text{th}} \in (0,1)$ such that for any $p < p_{\text{th}}$ and any bounded Hermitian operator X we have:

$$\mathbf{Tr} X \rho^{(t)} \leq \mathbf{Tr} X \mathcal{N}_p(\rho) + t||X||e^{-\Theta(\sqrt{n})}$$

Proof. By induction on t.

$$\begin{aligned}
\mathbf{Tr} X \rho^{(t)} &= \mathbf{Tr} X D \circ \mathcal{N}_p \rho^{(t-1)} \\
&\leq \mathbf{Tr} X D \circ \mathcal{N}_p \circ \mathcal{N}_p(\rho) + (t-1)||X||e^{-\Theta(\sqrt{n})} \\
&\leq \mathbf{Tr} X \mathcal{N}_p(\rho) + t||X||e^{-\Theta(\sqrt{n})}
\end{aligned}$$

When in the last transition we used Lemma 1.1. □